

ESG × AI Readiness Diagnostic

Scientific & Methodological Foundation Note

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This note explains **why the diagnostic is structured as it is** — its maturity model basis, 5-point scale, five dimensions, question count, weighting logic, and scoring formula — so that clients can evaluate the tool's validity with confidence.

1. Maturity Model Foundation

The diagnostic adapts the **Capability Maturity Model (CMM / CMMI)**, developed at Carnegie Mellon's SEI, to the ESG + AI domain. CMM's insight is that organisational capability evolves through predictable, qualitatively distinct stages — which is exactly what a readiness diagnostic must capture.

The five maturity levels:

- **1 — Ad hoc:** manual, fragmented, no AI.
- **2 — Initial:** basic tooling, siloed processes.
- **3 — Defined:** standardised, repeatable workflows.
- **4 — Managed:** data-driven and AI-augmented.
- **5 — Leading:** real-time, integrated, audit-ready.

A maturity scale produces **actionable remediation guidance** ("you are at Level 2 — build structured data processes next") that a binary pass/fail tool cannot.

2. The 5-Point Likert Scale

Each question uses a **behaviourally-anchored 5-point ordinal scale**. This design reflects three principles from psychometric research:

- **Optimal granularity:** Fewer than 5 levels loses discriminatory power; more than 7 does not add reliability and increases respondent burden (Likert 1932; Preston & Colman 2000).
- **Behavioural anchoring (BARS):** Each option describes a specific observable organisational state, not a vague numeric rating. This eliminates ambiguity and reduces social desirability bias.
- **Interval approximation:** 5-point symmetric labelled scales are widely accepted as approximating interval-level measurement, which justifies the weighted summation in the scoring formula (Norman 2010).

Example — Q7 Carbon Accounting

Level 1: No structured carbon accounting; emissions estimated ad hoc if at all.

Level 3: Scope 1 & 2 calculated from metered data; Scope 3 partially addressed.

Level 5: Full Scope 1–3 with AI-assisted dynamic updating and supplier data integration.

3. The Five Dimensions — Why These Five?

Dimensions were derived by clustering 35 documented ESG + AI use cases (drawn from PositionGreen, SustainGRC, IFRS/ISSB, EFRAG, UBP, OECD, Catalyst, Cornell Tech, SSIR, and DIESIS Network) into groups sharing a **common data infrastructure requirement, governance challenge, and independent diagnostic value**. The result mirrors the functional architecture of a modern ESG programme — not the E/S/G reporting taxonomy.

Dimension	Qs	Max	Why this many questions & points?
Data & Reporting Infrastructure	3	23 pts	Upstream dependency — all other AI use cases require structured, validated data.
Compliance & Risk	3	23 pts	Governs auditability and assurance-readiness under mandatory CSRD limited assurance.
Environmental Impact	3	22 pts	Most data-intensive dimension; Scope 3 and land-use use cases anchor this cluster.
Social & Stakeholder Intelligence	2	15 pts	Fewer but high-stakes use cases; dual-use risk requires governance calibration.
Governance & Investment	2	15 pts	Highest-order capabilities — integrate outputs of all other dimensions.

The first two dimensions carry more aggregate weight because they are **upstream dependencies**: no ESG AI use case is viable without structured, auditable data. Adding a 6th dimension would fragment the environmental cluster without diagnostic gain; collapsing to 4 would merge data collection with compliance governance — two distinct readiness problems.

4. Question Weighting

Every question carries a weight of either **7 or 8 points** (total: 98, normalised to 100). This narrow range enforces the **anti-dominance principle**: no single question exceeds ~8% of the total, so no single strong answer can mask systemic weakness elsewhere.

- **8-point questions (7 of 13)**: anchor use cases — highest regulatory exposure and broadest applicability within each dimension.
- **7-point questions (6 of 13)**: supporting use cases — important but slightly more specialised.

Equal weights were evaluated and rejected: a question about automated data aggregation (without which no other ESG AI works) deserves at least as much weight as one about biodiversity supply-chain monitoring. The 7/8 spread operationalises this priority difference while preserving the anti-dominance constraint.

5. Scoring Formula & Aggregation

$$\text{Points}(q) = (\text{Response} \div 5) \times \text{Weight}$$

$$\text{Total} = \text{round}(\Sigma \text{ Points} \div 98 \times 100)$$

The formula is linear, transparent, and fully reproducible. A perfect score on every question = 100; a minimum score on every question ≈ 20 (preserving a non-zero floor). Dimension scores use the same formula restricted to that dimension's questions.

Maturity bands:

- **85–100**: High Readiness
- **65–84**: Moderate Readiness
- **40–64**: Limited Readiness
- **0–39**: Critical Gap

Thresholds are calibrated against the KPMG ESG Assurance Maturity Index (2023), CDP A-List analysis (2024), and Conference Board S&P 500 ESG data (2024).

6. Answers to Common Client Questions

Why 13 questions?

13 is the minimum to cover all five dimensions with sufficient granularity to distinguish maturity levels within each. Every question maps to a specific documented use case — there are no fillers. Adding questions would introduce lower-priority (Tier 2) use cases without material diagnostic gain; removing any would leave a dimension with single-item coverage. The set is optimised for 6–8 minute completion to prevent fatigue-induced response degradation.

Why 5 dimensions — not E/S/G?

E/S/G are disclosure categories, not readiness dimensions. Data & Reporting Infrastructure and Compliance & Risk have no equivalent in the E/S/G taxonomy but are the two most critical determinants of AI readiness. The five dimensions map to how ESG programmes actually function, not how they report.

Why weights of 7 and 8?

The 7/8 range satisfies three constraints: (1) total normalises cleanly to ~100; (2) spread is narrow enough to enforce anti-dominance (~12.5% difference); (3) within-dimension priority is encoded without any question dominating the result. A 6/10 spread, for example, would allow a single 10-point question to represent >10% of the total.

Is this statistically validated?

The tool applies deductive construct validity: every question is derived from a sourced use case, every dimension is grounded in a coherent theoretical construct. The formula is fully transparent. Psychometric refinement (Cronbach's alpha) is planned as the respondent dataset grows.

Can a strong dimension offset a weak one?

No — by design. The maximum any single dimension can contribute is 23 pts. An organisation scoring 5/5 on Governance while scoring 1/5 on Data & Reporting will still fall in the Critical Gap band. Systemic weakness cannot be masked by isolated strength.